Ismail Gul

PhD Candidate, Stevens Institute of Technology
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RESEARCH INTERESTS

Remote Sensing, Hydrometeorology, Weather Radar Signal Processing, Precipitation Nowcasting, Internet of Things, Flood Monitoring, Quantitative Precipitation Estimation, and Environmental Sensing

EDUCATION

2024–Present *(Expected: 2027)*
Ph.D., Civil, Environmental and Ocean Engineering **GPA: 4.00/4.00**
Stevens Institute of Technology – NJ, USA
Dissertation Title: Enhancing Urban Flood Monitoring through Improved Observational Data
Advisor: Marouane Temimi

2018–2021
M.S., Electrical and Electronics Engineering
Istanbul Technical University – Istanbul, Turkey
Thesis Title: Learning Scanning Regime in Electronic Support Receivers Using Matrix Decompositions
Advisor: Isin Erer

2012–2016
B.S., Electrical and Electronics Engineering
Bilkent University – Ankara, Turkey
Graduation Project: Target Detection, Tracking and Depth Estimation Using Long Span Stereo Camera System

PUBLICATIONS

Peer-Reviewed Journal Articles

1. **Gul, I.**, Temimi, M., and Chandrasekar, V. (2026). "Enhancing Precipitation Nowcasting Guided by Physics-Based Predictions." *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*. doi:10.1109/JSTARS.2026.3746059.
2. **Gul, I.**, Abdelkader, M., Mackelburg, K., and Temimi, M. (2025). "Dual-Purpose and Low-Cost IoT-Based Sensor Network for Real-Time Rainfall and Urban Flood Monitoring." *IEEE Internet of Things Journal*. doi:10.1109/JIOT.2025.3636404.
3. **Gul, I.** and Erer, I. (2022). "Randomized Low Rank and Sparse Decomposition Based Receiver Search Strategy for Electronic Support Receivers." *IEEE Transactions on Aerospace and Electronic Systems*. doi:10.1109/TAES.2022.3200646.

Peer-Reviewed Conference Papers

1. **Gul, I.** and Erer, I. (2021). "Learning Scanning Regime for Electronic Support Receivers by Nonnegative Matrix Factorization." *44th International Conference on Telecommunications and Signal Processing (TSP)*, pp. 231–234. doi:10.1109/TSP52935.2021.9522658.
2. **Gul, I.** and Erer, I. (2021). "Scanning Strategy Learning for Electronic Support Receivers by Robust Principal Component Analysis." *Artificial Intelligence and Machine Learning in Defense Applications III*, SPIE, Vol. 11870, Article 118700Q. doi:10.1117/12.2601109.

Manuscripts in Preparation

1. **Gul, I.** et al. "Evaluation of X-Band Weather Radar Precipitation Estimates Using Operational Radar Products and Surface Observations" *Manuscript in preparation.*

CONFERENCE PRESENTATIONS

Mar 2026 **Gul, I.** and Temimi, M. "Stevens High-resolution Atmospheric Radar Project (SHARP)." *NSF NCAR Radar Technology Community Workshop*. Poster Presentation. Poster.

Dec 2025 **Gul, I.**, Gutierrez Diaz, S., Udenze, P., Balthasar, J., Davis, L., Li, D., Wilson, C., and Lotspeich, R. R. "A Low-Cost and Rapid Deployment Flood Level Observation System." *AGU Fall Meeting 2025*. Poster Presentation. Abstract.

May 2025 **Gul, I.** and Temimi, M. "Advancing the Monitoring of Pluvial Flood Inundation Using Low-Cost IoT-based Rainfall and Water Level Sensor Network." *CIROH Training and Developers Conference 2025*. Poster Presentation. Abstract.

Dec 2024 **Gul, I.**. "IoT-Based Sensor Network for Real-Time Rainfall and Urban Flood Monitoring." *AGU Fall Meeting 2024*. iPoster Presentation. iPoster.

PROFESSIONAL EXPERIENCE

ASELSAN Inc. – Radar and Electronic Warfare Division Ankara, Turkey

2023–2024 **Lead System Engineer**

2021–2023 **Senior System Engineer**

2016–2021 **System Engineer**

Contributed to radar and electronic warfare projects throughout the system lifecycle, from conceptual design and requirements engineering to system integration, verification, and acceptance.

TEACHING EXPERIENCE

Spring 2026 **Teaching Assistant**, CE 304: Water Resources Engineering
Stevens Institute of Technology

Spring 2025 **Teaching Assistant**, CE 304: Water Resources Engineering
Stevens Institute of Technology

2025–Present **Research Mentor**, Undergraduate Student
Robert Galetta – Stevens Institute of Technology

2024–2025 **Research Mentor**, Undergraduate Student
Kyle Mackelburg – Stevens Institute of Technology

HONORS AND AWARDS

2026 **NSF NCAR Radar Technology Community Workshop Travel Grant**
NSF National Center for Atmospheric Research (NCAR)

2025 **USGS FLOW Academy Fellowship**
USGS and CIROH

2024 **AGU Michael H. Freilich Student Visualization Competition**
American Geophysical Union (AGU)

2024 **Provost Doctoral Fellowship**
Stevens Institute of Technology

2021 **Academic Excellence Certificate**
ASELSAN Academy

2012–2016 **Honor and High Honor Certificates**
Bilkent University

TECHNICAL SKILLS

Programming: Python, MATLAB, Arduino, JavaScript, HTML, Node.js, MongoDB, C#, C++, Java, OpenCV.
Software: PyTorch, TensorFlow, CUDA, pySWMM, Android Studio, JIRA, IBM DOORS and IBM ClearQuest.
Hardware: Microcontroller, sensor integration and calibration, 3D printing and prototyping, testing and debugging.

PROFESSIONAL SERVICE

2026–Present	**Peer Reviewer** IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS)
2025–Present	**Peer Reviewer** IEEE Transactions on Aerospace and Electronic Systems (TAES)
Aug 2025	**Student Volunteer** 41st International Conference on Radar Meteorology, American Meteorological Society (AMS), Toronto, Canada
Dec 2024	**Student Volunteer** AGU Fall Meeting 2024, American Geophysical Union (AGU), Washington, DC, USA